

TDC 2026 Final Report

TTC Bus and Streetcar Delay Analysis

The Delayticans

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Abstract

This report summarizes our work on the TDC 2026 project analyzing Toronto Transit Commission (TTC) bus and streetcar delay data. We merged multi-year CSV datasets, built Python tooling to combine and visualize the data, and produced charts that highlight rows with zero or null delay and gap values across vehicle types and years.

1 Introduction

We are submitting our final report for the TDC 2026 project. Our team, The Delayticans, analyzed publicly available TTC delay records for buses and streetcars. The goal was to consolidate fragmented yearly datasets into unified files and explore data quality patterns that may affect downstream analysis.

2 Data Sources

Raw CSV files are stored under `raw_data_sources/`, organized by vehicle type:

- **Buses** — annual delay files from 2014 through 2024, plus a combined file for 2025 onward
- **Streetcars** — annual delay files from 2014 through 2024, plus a combined file for 2025 onward

Each dataset includes delay-related fields such as `Min Delay` and `Min Gap`. Code description files are provided alongside the raw data for reference.

3 Methodology

3.1 CSV Merging

We implemented `scripts/merge_csv.py` to concatenate multiple CSV files in order. The script:

- writes a single header row from the first input file
- skips duplicate headers in subsequent files
- warns when headers differ between files

A companion runner, `scripts/merge_csv_run.py`, allows merging from the editor without using the terminal. This workflow was used to combine monthly streetcar files into yearly datasets and to prepare unified bus and streetcar files for analysis.

3.2 Data Visualization

We implemented `scripts/data_visualiser.py` to generate charts from the merged datasets. For each CSV file, the script counts rows where `Min Delay` or `Min Gap` is zero or null and produces:

- a pie chart showing the proportion of affected rows versus all other rows
- a yearly bar chart summarizing the percentage of zero/null rows by vehicle type

Output figures are saved under `data_visualisations/Buses/` and `data_visualisations/Streetcars/`.

4 Completed Work

We completed the following tasks for this submission:

1. Merged CSV files for TTC bus and streetcar delay data across multiple years.
2. Built Python scripts to automate merging and visualization.
3. Generated per-year pie charts and yearly summary bar charts for both vehicle types.
4. Prepared this technical report documenting our data pipeline and findings.

5 Results

Figures ?? and ?? show the share of rows with zero or null delay/gap values by year. Individual pie charts for each source file are available in the project repository under `data_visualisations/`.

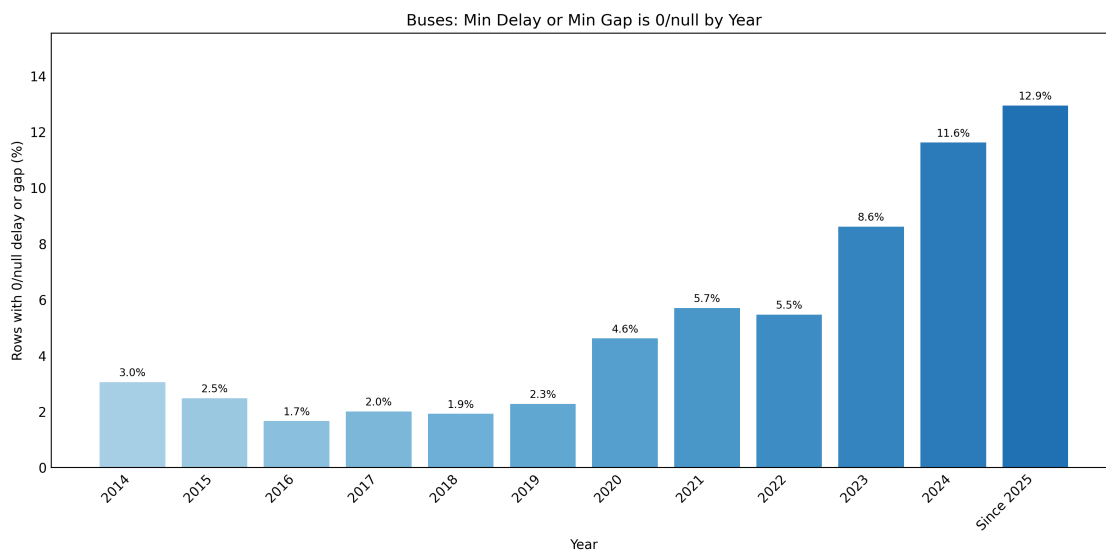


Figure 1: Buses: percentage of rows with zero or null delay or gap by year.

6 Conclusion

We consolidated TTC bus and streetcar delay datasets, automated CSV merging, and produced visualizations that highlight data-quality patterns across years. This work provides a reusable foundation for further analysis, including geographic mapping and deeper delay trend studies.

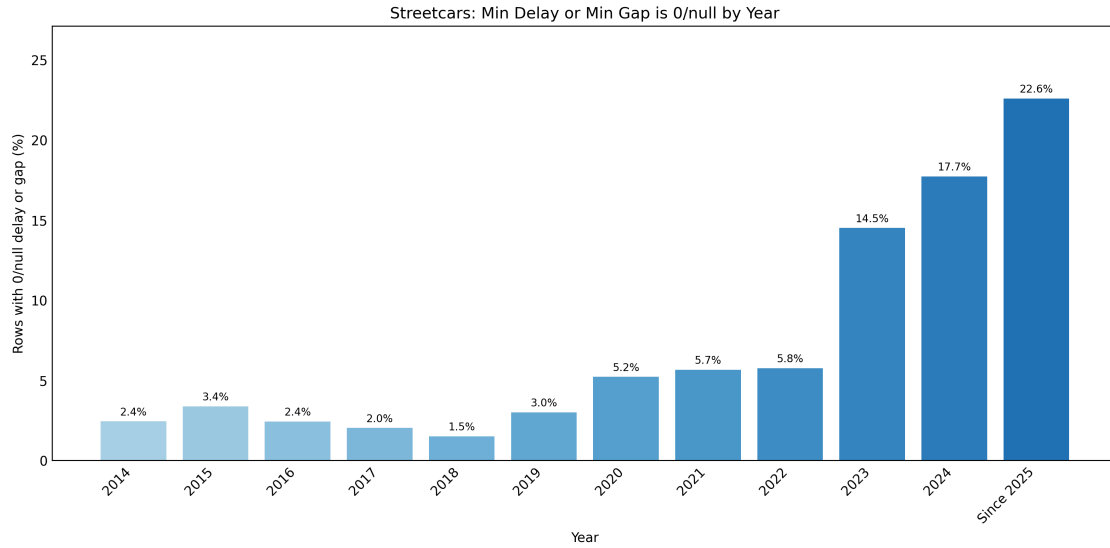


Figure 2: Streetcars: percentage of rows with zero or null delay or gap by year.

7 Repository Structure

Path	Description
<code>raw_data_sources/</code>	Original TTC CSV files
<code>scripts/merge_csv.py</code>	CSV merge utility
<code>scripts/data_visualiser.py</code>	Chart generation script
<code>data_visualisations/</code>	Generated PNG figures